

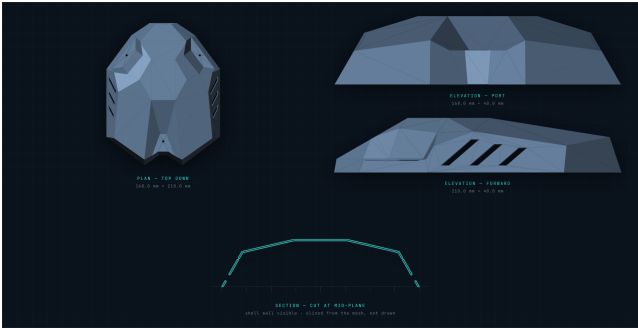
SeaNergy Enclosure Drawings

V1 evidence geometry and V2 pressure-housing targets | Dimensions in mm

V1 MESH - NOT WATERPROOF



V2 FINISHED-PRODUCT VIEW | GRAPHITE HOUSING AND SERVICE ACCESS CAP



Property	Calculated / inspected value	
Bounding envelope	168.0 x 210.0 x 40.0 mm	Confirmed
Mesh	3,934 facets; 77,771.9 mm2 surface	Geometry
Topology	49 edges do not have incidence 2	Not a close
Signed mesh volume	45,825.7 mm3	Unreliable
Solid PETG mass from mesh volume	56.8 g at 1.24 g/cm3	Material es
Tolerance	V1 not dimensioned	V2 drawing

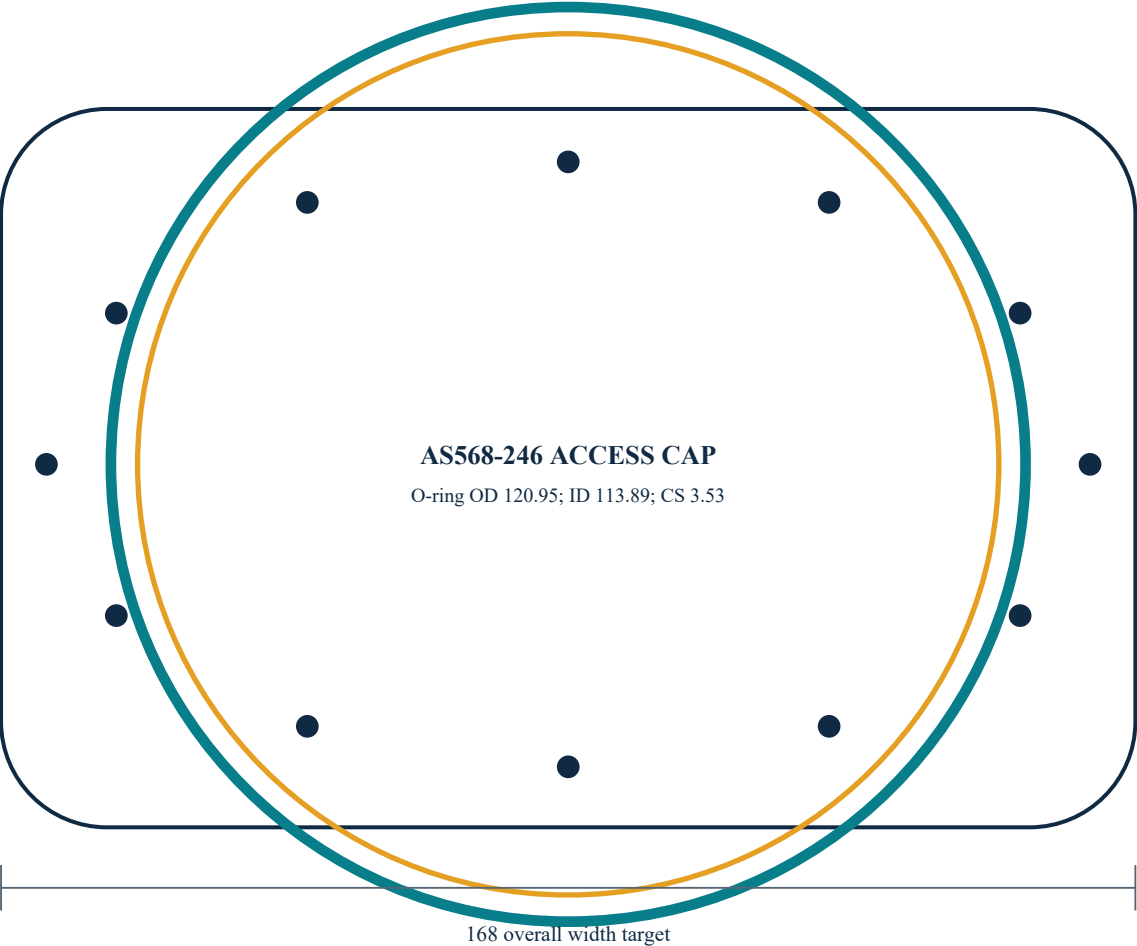
Current evidence: the V1 form demonstrates AUV fit intent and a real 168 x 210 x 40 mm envelope. It does not yet provide a lid, controlled seal land, penetrator bore or pressure rating.

Source: DAA Hyper Grey.stl and housing renders. Topology was calculated directly from the binary STL triangles.

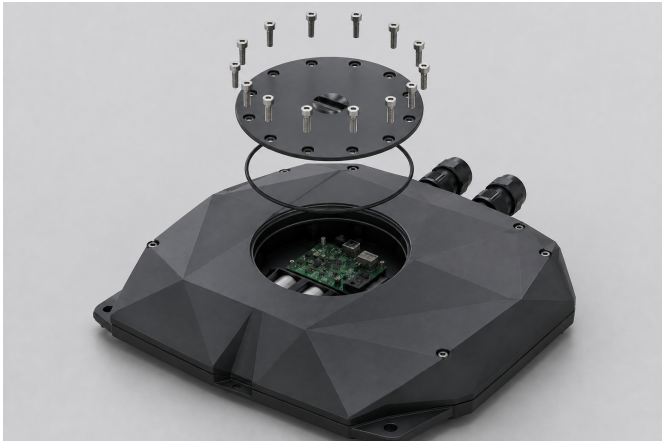
SeaNergy Enclosure Drawings

V2 access-cap seal and fastening system | Preliminary - design review required

V2 DESIGN TARGET



Seal parameter	Nominal	Tolerance / calculation
O-ring	AS568-246 NBR 70	113.89 +/-0.76 ID; 3.53 +/-0.10 CS
Face groove depth	2.82 mm	+0.00/-0.05; 20.1% nominal axial squeeze
Face groove width	4.80 mm	+0.10/-0.00; 72.3% nominal gland fill
Groove OD control	121.00 mm	External pressure seats ring against OD
Seal finish	Ra <=0.8 um target	Printed prototype must be machined or resin-sealed
Fasteners	12 x M3 A4-70	0.50 N m target in cross pattern
Penetrators	2 x WetLink M10 target	Final pressure rating from selected manufacturer



Exploded assembly view | Seal stack and fastening arrangement

Dimensions are design targets, not as-built inspection results. Do not pressure test a printed part without a shielded test plan.

SeaNergy Enclosure Drawings

Mass, displacement, AUV fit and verification ledger

ALL WATER VALUES ESTIMATED

No.	Property	Current V1	Finished-product target	Verification
M-01	External envelope	168 x 210 x 40 mm STL	Same maximum fit envelope	Callipers and hull-section gauge
M-02	Shell mass	56.8 g PETG calculation from open mesh volume	<=350 g sealed V2 shell and cap	Scale, 1 g resolution
M-03	Complete mass in air	Not assembled	1.25 kg planning target	Scale after cable and battery fit
M-04	Displaced water	Cannot infer from open mesh	1.20 L target	Archimedes displacement test
M-05	Apparent mass in freshwater	Not measured	0.05 kg downward from 1.25-1.20 kg	Suspended scale in 20 degC water
M-06	Dunk	0 tests	1 m equivalent, 60 min, dry tissue	Photographs and pre/post mass
M-07	Pressure	No rating	30 bar design objective; staged proof defined later	Shielded hydrostatic chamber
M-08	AUV fit	Rendered envelope only	2 mm minimum hull clearance	Printed hull-section go/no-go gauge

Achievement path: repair V2 CAD to a closed solid; print and inspect the seal land; fit the standard O-ring and pressure-rated penetrators; pass 10 cm/10 min, 1 m/60 min and shielded pressure stages; then record mass, displacement and hull clearance. Until these gates pass, the enclosure remains a fit prototype.